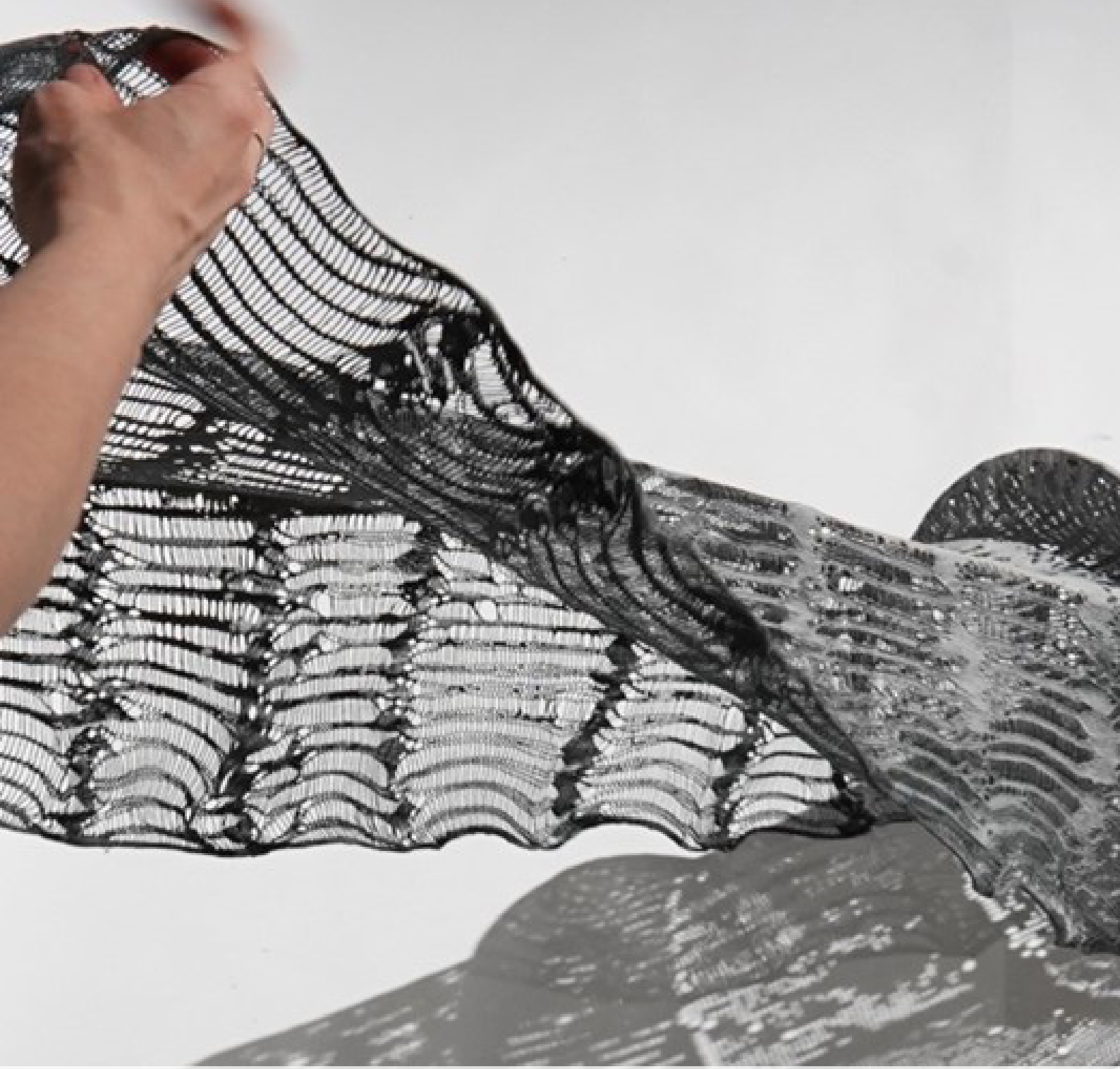
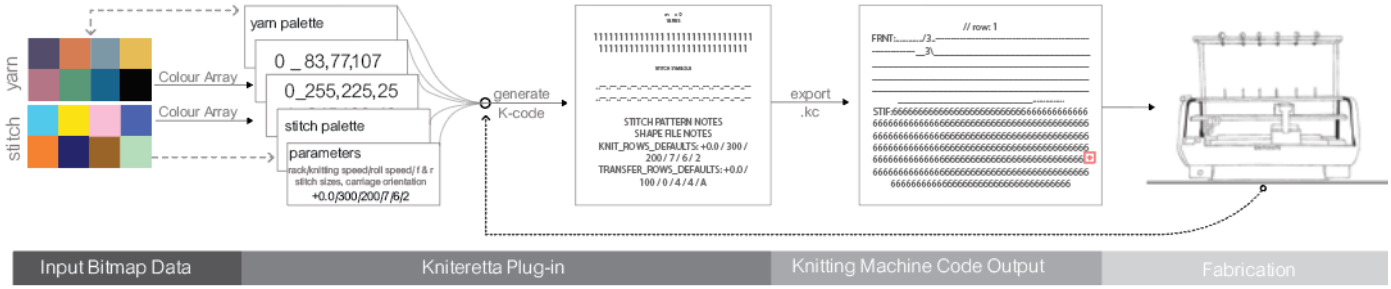


User Guide

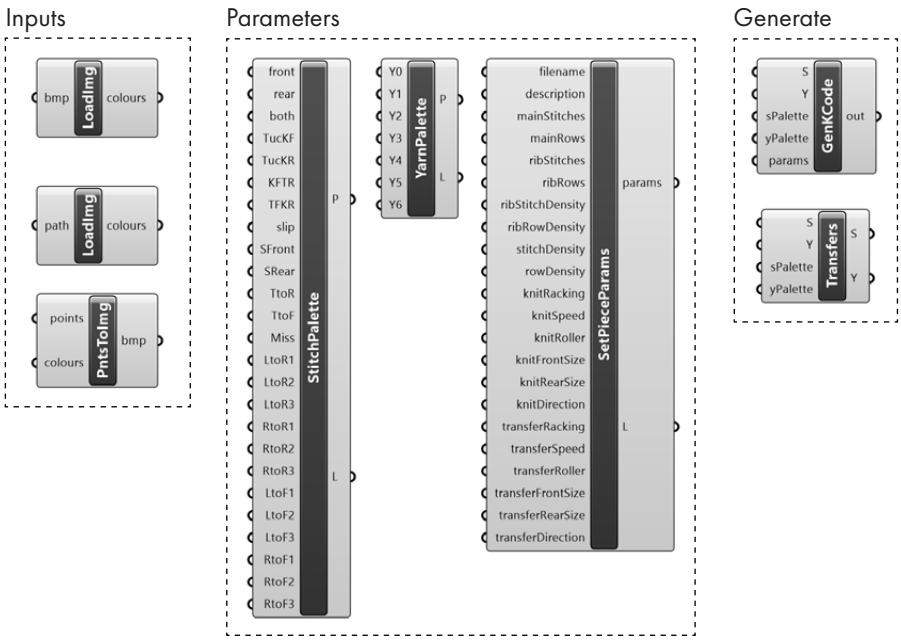
Kniteretta

Kniteretta converts bitmap images into machine-readable text files, making the design process less reliant on specialized knitting software.

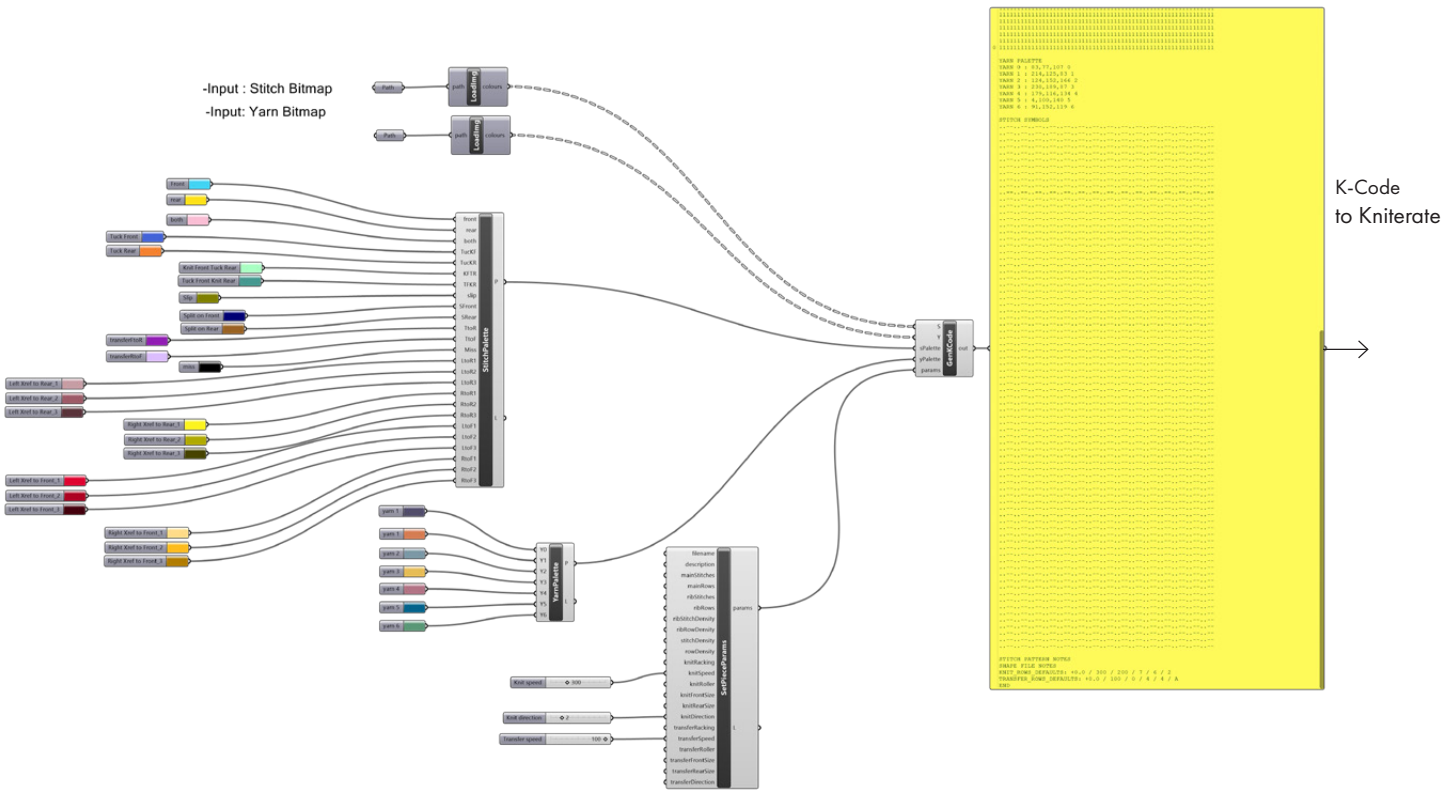
WORKFLOW



COMPONENTS



INTERFACE



Inputs

LoadImg

path to colours - Converts a bitmap image into a colour array. The component reads each pixel in the image, identifies its colour, and assigns an RGB value. Each pixel corresponds to a single stitch, and its colour can define the stitch type or the yarn type.

bmp to colours - Converts a bitmap image into a bidimensional array (tree) of the colours read, where colours can be used to define a stitch type or yarn selection.

PntsToImg

point clouds to bmp - Converts collections of points to a bitmap image. Points must be grouped, with each group representing a specific colour. The points should be scaled to a 1×1 m grid, where one point equals one pixel in the final image.

Parameters

StitchPalette*

Creates a colour palette that matches RGB colours with stitch types and needle movements.



*Note: The RGB values in the Palettes must match the RGB values used to create the bitmap.

YarnPalette*

Creates a colour palette that matches RGB colours to yarn feeders in Kniterate, using values from 0–6 for the six yarns, with 0 indicating no yarn.



SetPieceParams

Allows modification of properties such as rack (movement of the knitting bed), knitting speed, roll speed, front and rear stitch sizes, and carriage orientation.
Note: It is also possible to overwrite and apply custom values to individual rows on the exported Text file.

Generate

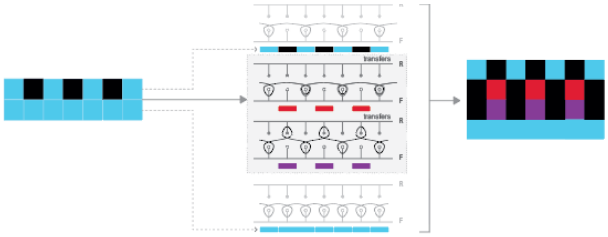
Generate K-Code

Generates K-Code from stitch and yarn data. This text should be saved in .txt format.

Transfers**

Detects and automatically creates 'transfer pixels' for the input images. This must be done for both the stitch file and the yarn file.

**Note: currently works on 1x1 incremental transfers

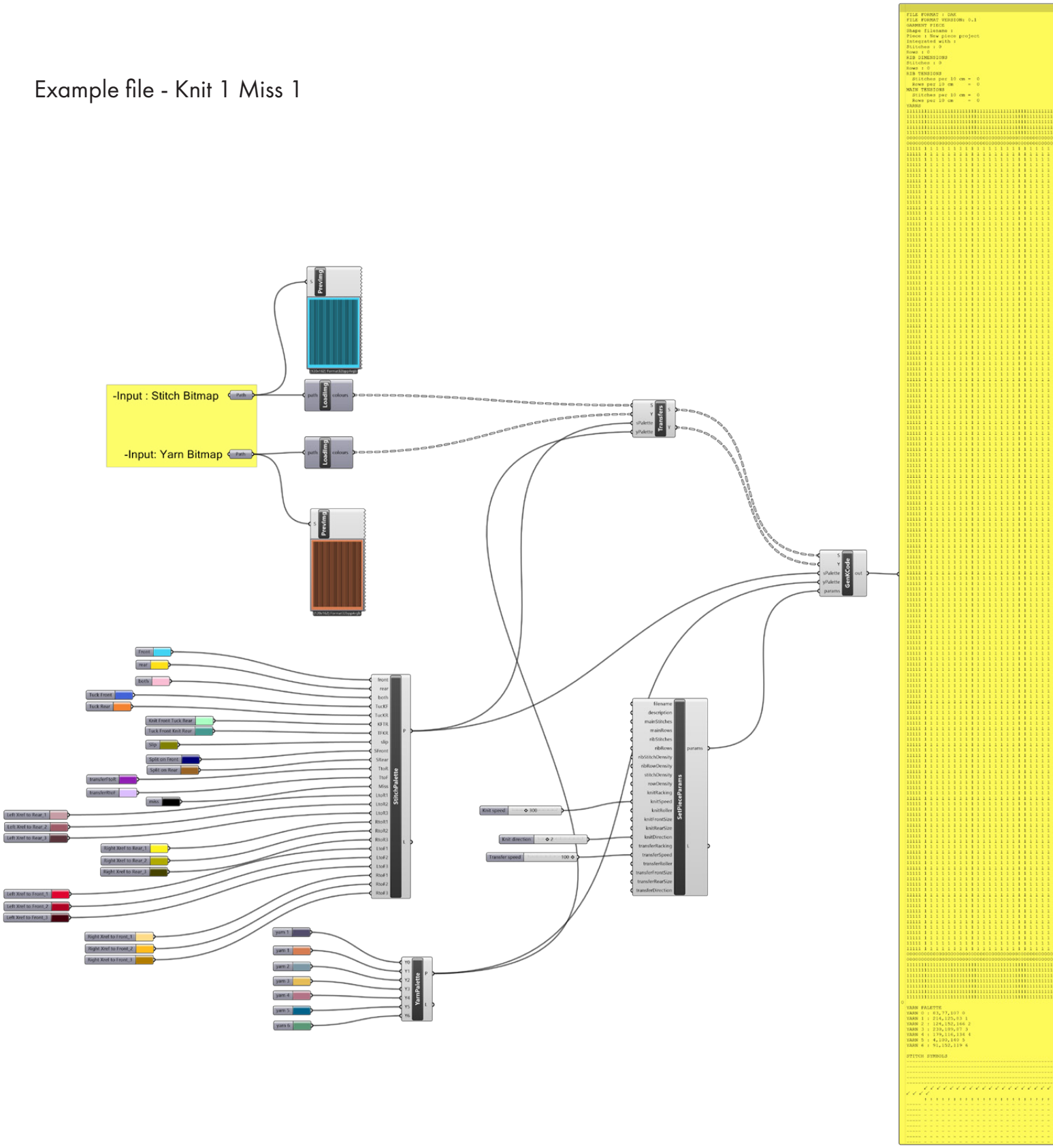


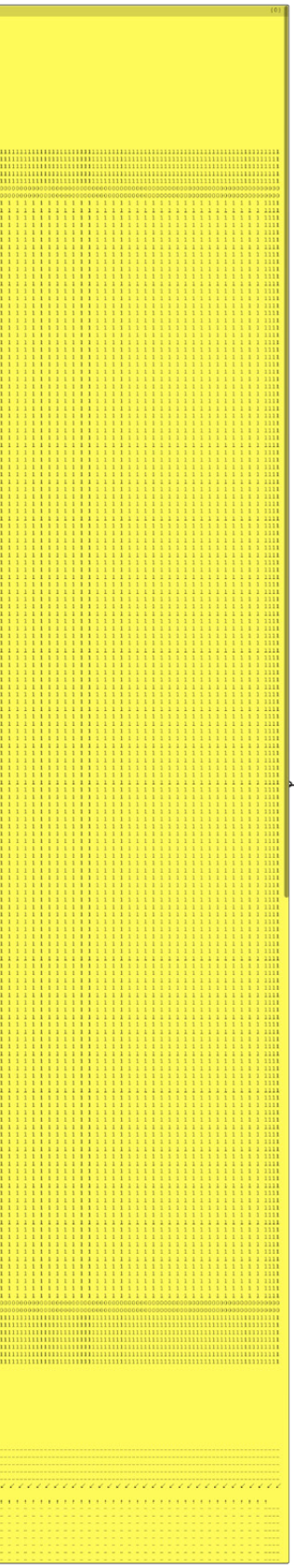
Export to Kniterate

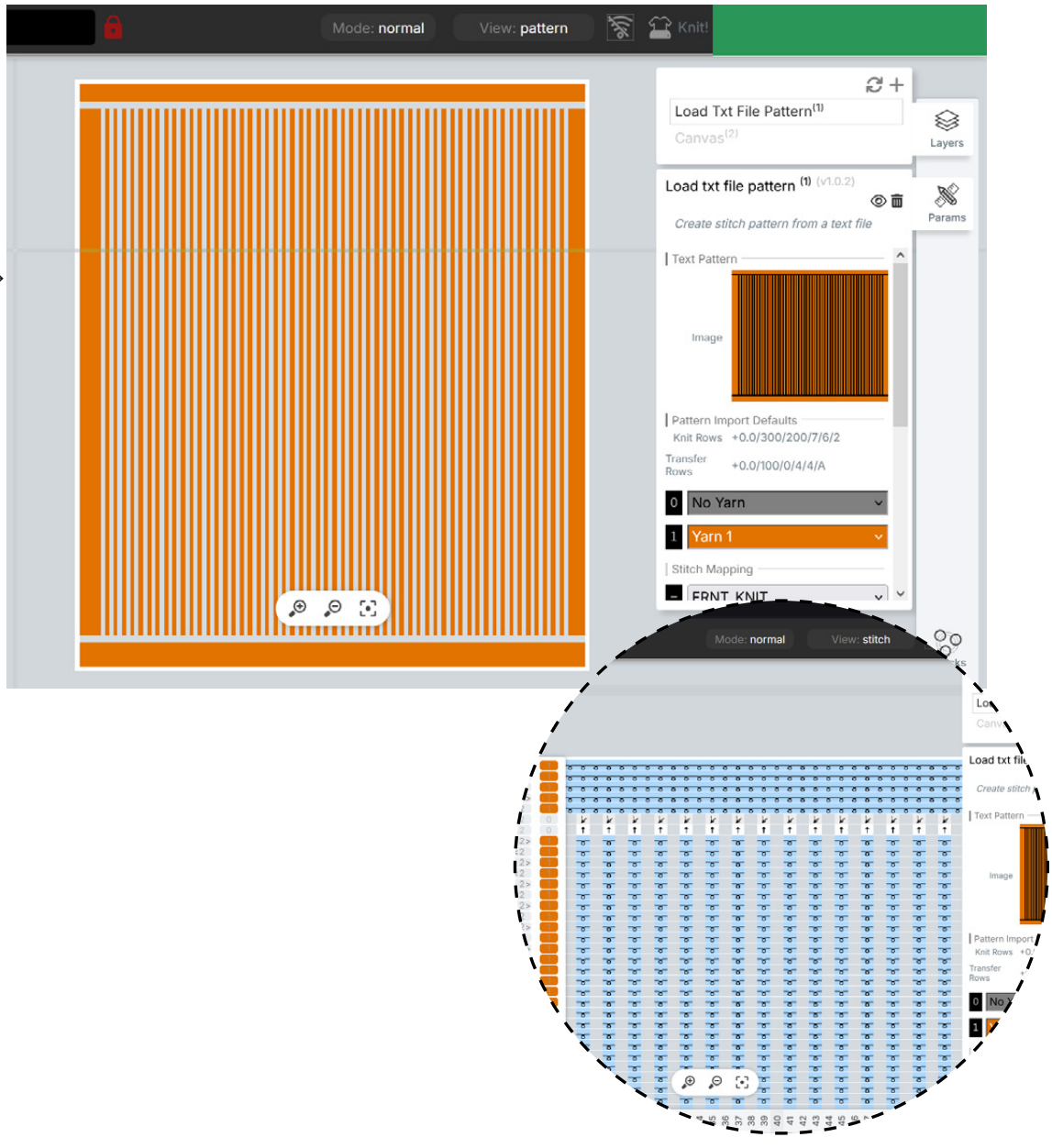
Load Text pattern

Load the saved K-Code on the Kniterate using the "Load Text File Pattern" function.

Example file - Knit 1 Miss 1






Mode: normal
View: pattern
Knit!
Load Txt File Pattern(1)
Canvas(2)
Layers
Load txt file pattern (1) (v1.0.2)
Create stitch pattern from a text file
Params
Text Pattern
Image
Pattern Import Defaults
Knit Rows +0.0/300/200/7/6/2
Transfer Rows +0.0/100/0/4/4/A
0 No Yarn
1 Yarn 1
Stitch Mapping
FRNT_KNIT
Mode: normal
View: stitch

Alternative ways of generating bitmap on grasshopper

